FINANCE ANALYST PROJECT | DATA ANALYTICS PORTFOLIO

Supply Chain Management Demand Forecasting & Operational Analysis

Presented by Pinkal Patel

Technologies: Python • Machine Learning • Pandas • Scikit-learn • XGBoost

BUSINESS PROBLEM

Modern supply chains generate large volumes of operational data across suppliers, logistics, production, and sales. However, many organizations struggle to transform this data into clear operational insights.

- Supplier risks remain unnoticed
- Inventory inefficiencies cause capital lock-in
- Transportation delays affect customer satisfaction
- Operational costs reduce profit margins

- The lack of centralized visibility leads to slow and inefficient decision making

PROJECT GOALS

01 Monitor supplier reliability and defect rates

02 Analyze inventory investment and stock turnover

03 Identify high-performing and low-performing SKUs

04 Understand revenue contribution across product categories

05 Analyze demand distribution across products

PROJECT OBJECTIVES

01 Build a machine learning demand forecasting system

02 Analyze supply chain dataset

03 Transform raw data into ML features

04 Train predictive models

05 Evaluate model performance

BUSINESS IMPACT

Improved Demand Forecasting: Accurate prediction of product demand enables better planning and reduces uncertainty in supply chain operations

Optimized Inventory Management: Minimizes stockouts and overstock situations, leading to efficient inventory utilization and reduced holding costs

Cost Reduction: Helps in reducing operational costs related to storage, wastage, and emergency procurement

Efficient Supply Chain Operations: Enhances logistics planning, warehouse management, and distribution efficiency

Better Supplier Management: Supports data-driven decisions for selecting reliable and cost-effective suppliers

DATASET OVERVIEW

The dataset represents a fashion and beauty supply chain operation.

- **Product Information :** Product Type, SKU, Price, Revenue Generated
- **Inventory Data :** Stock Levels, Order Quantities, Inventory Value
- **Supplier Data :** Supplier Name, Lead Time, Defect Rate
- **Logistics Data :** Shipping Times, Shipping Carriers, Transportation Modes
- **Cost Data :** Manufacturing Cost, Shipping Cost, Total Cost

TECHNOLOGIES USED

Programming Language

- Python

Libraries & Frameworks

- Pandas – Data manipulation and preprocessing
- NumPy – Numerical computations
- Matplotlib & Seaborn – Data visualization
- Sit-learn – Machine learning models and evaluation
- XGBoost – Advanced boosting algorithm for high accuracy

Machine Learning Techniques

- Regression Models (Random Forest, Gradient Boosting, XGBoost)
- Time Series Analysis
- Feature Engineering & Hyperparameter Tuning

PROJECT WORKFLOW

Data Collection	Model Training & Tuning
Data Preprocessing	Model Evaluation
Exploratory Data Analysis (EDA)	Prediction & Forecasting
Feature Engineering	Visualization & Insights
Model Selection	Deployment / Output

DATA PREPROCESSING

Cleaning Steps:

- Removed missing values
- Standardized data formats
- Converted date columns
- Removed inconsistencies

Insight:

Clean data improves model accuracy and reliability

Handled missing and invalid records

Converted data into structured format

Prepared dataset for machine learning

EXPLORATORY DATA ANALYSIS

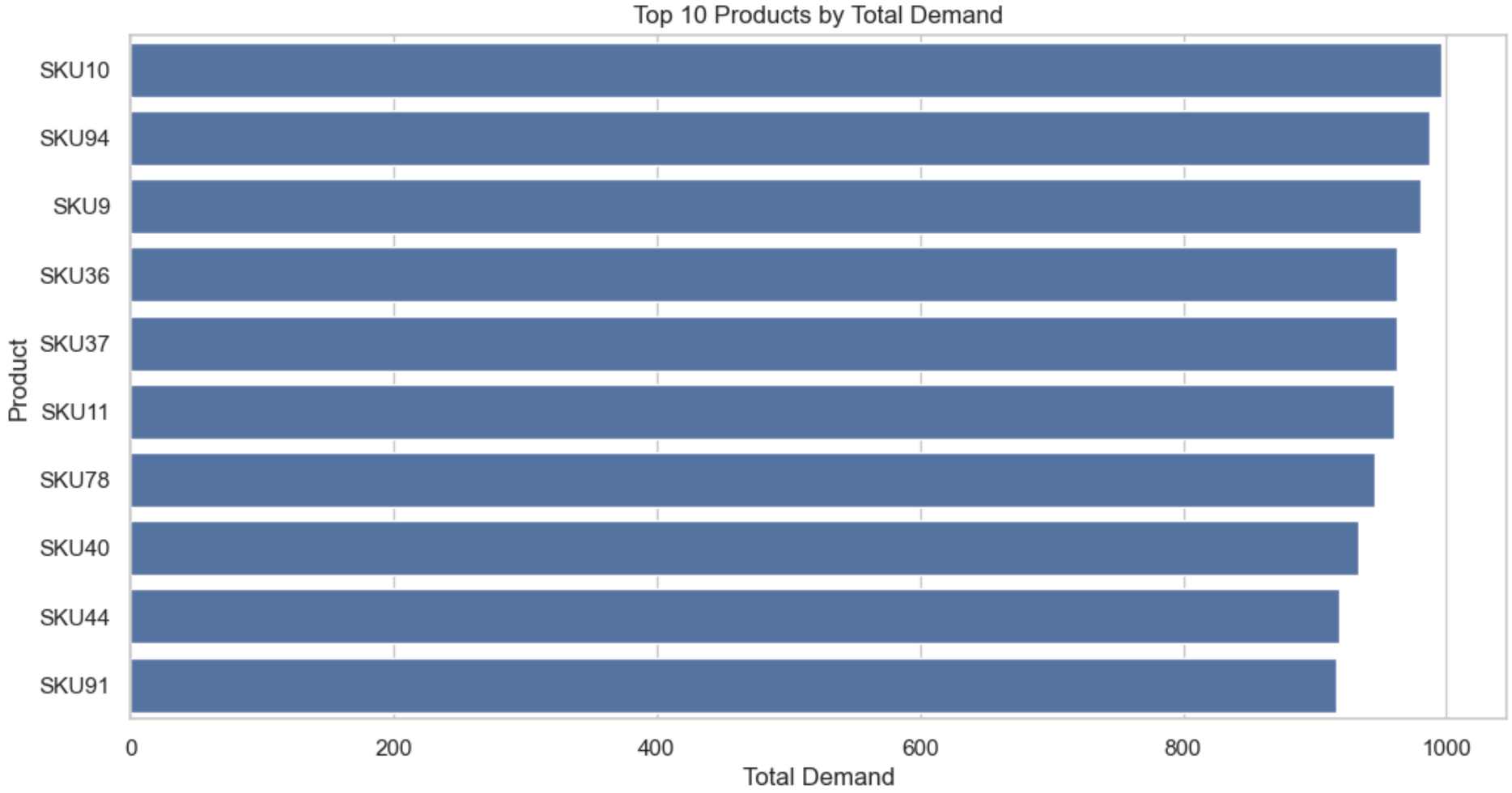
Key Questions:

- Which products have highest demand?
- Is demand evenly distributed?
- Are there dominant SKUs?

Purpose:

Understand patterns before building ML model

DEMAND DISTRIBUTION ANALYSIS



DEMAND DISTRIBUTION ANALYSIS

Insight:

- Demand is concentrated in a few top SKUs.
- These products drive most of the total demand.
- Prioritizing them helps avoid stockouts and improve efficiency.
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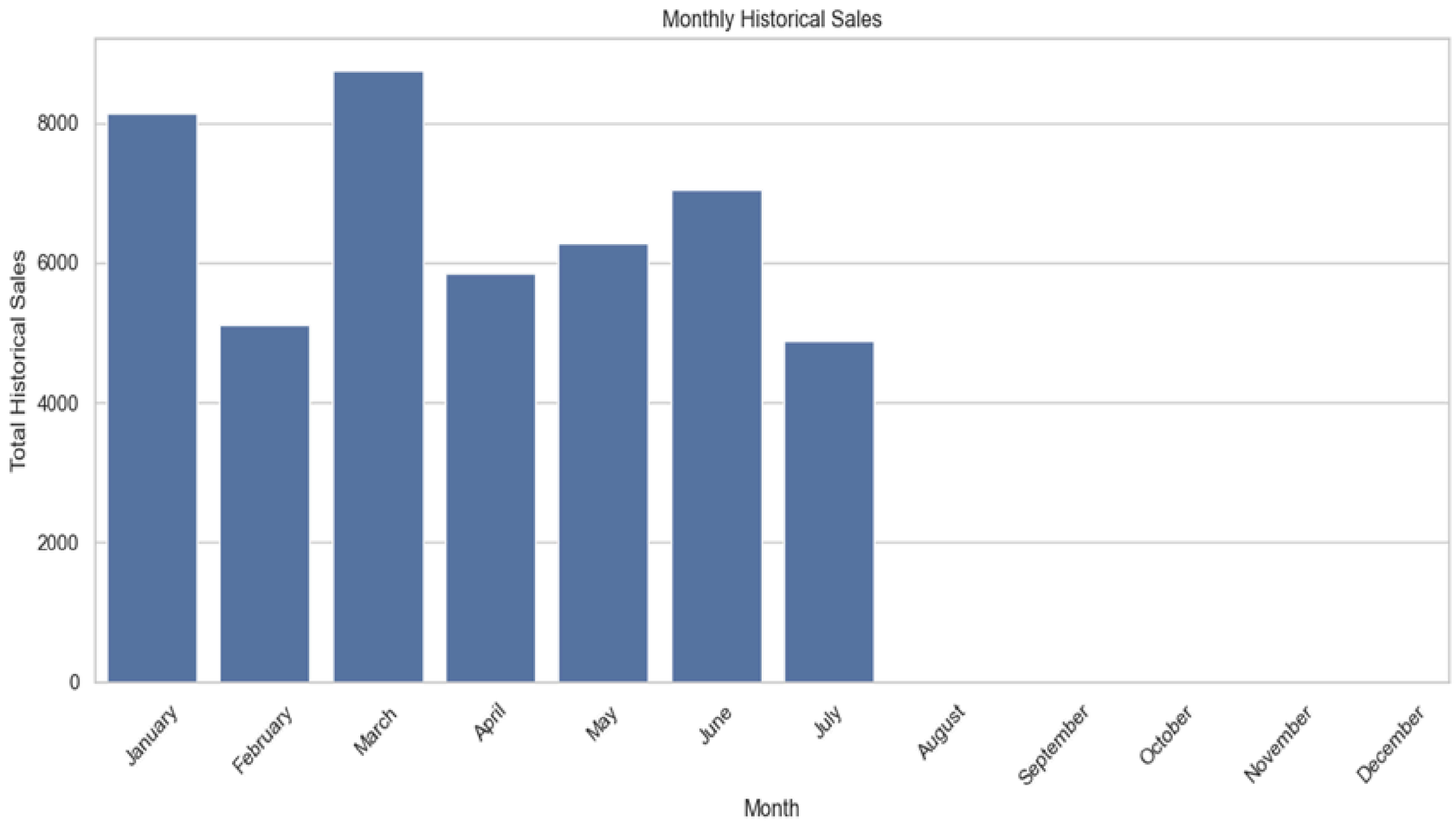
Business Value:

- Focus inventory on high-demand products
- Improves inventory planning
- Supports better resource allocation

SUPPLIER-WISE SALES PERFORMANCE



SUPPLIER-WISE SALES PERFORMANCE

Insight:

- Supplier 1 and Supplier 2 generate the highest sales
- Sales are moderately distributed among remaining suppliers
- Clear performance gap between top and low-performing suppliers
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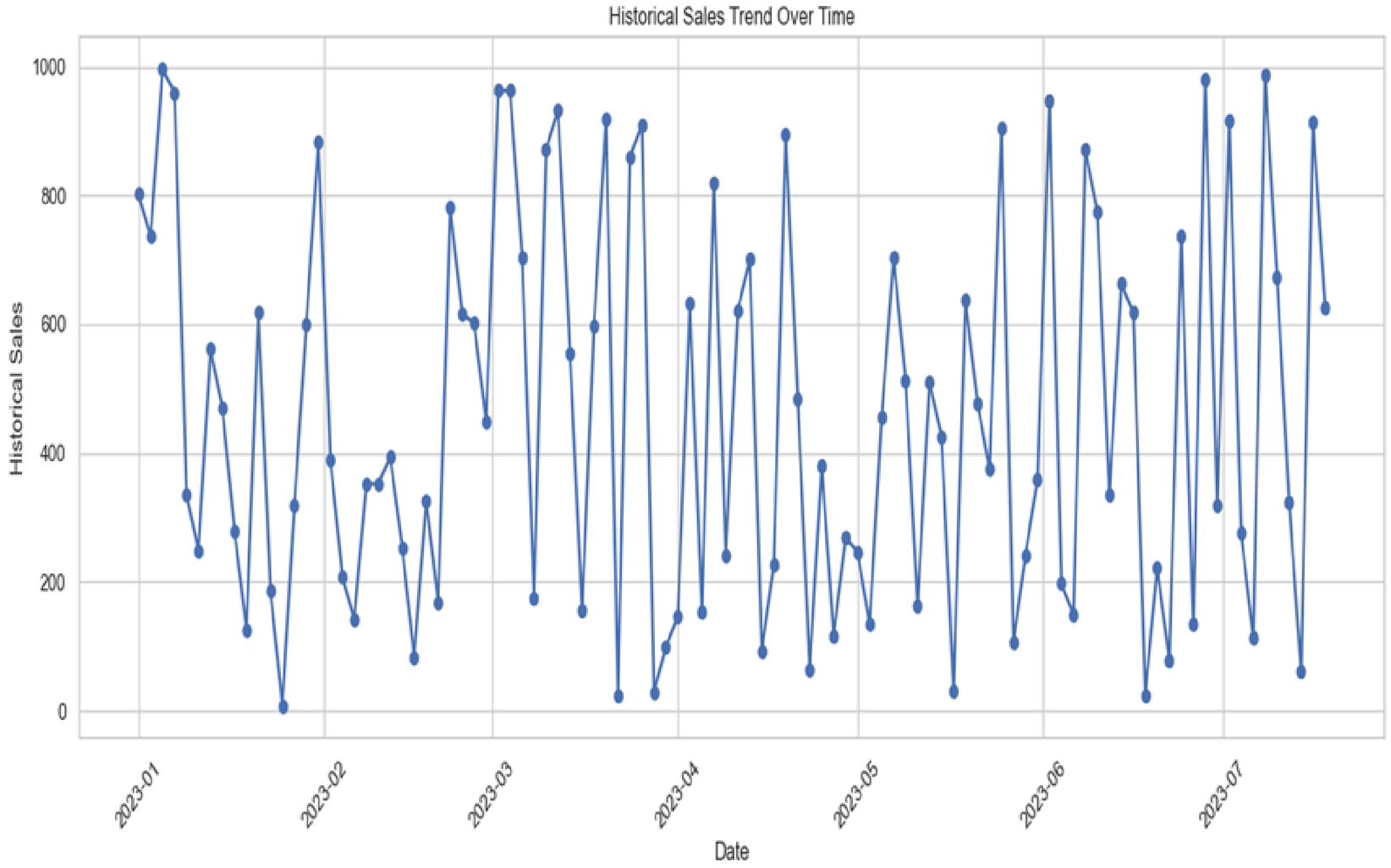
Business Value:

- Helps identify top-performing suppliers for strategic partnerships
- Supports better supplier allocation and demand planning
- Enables performance-based supplier optimization

HISTORICAL SALES TREND



Historical Sales Trend

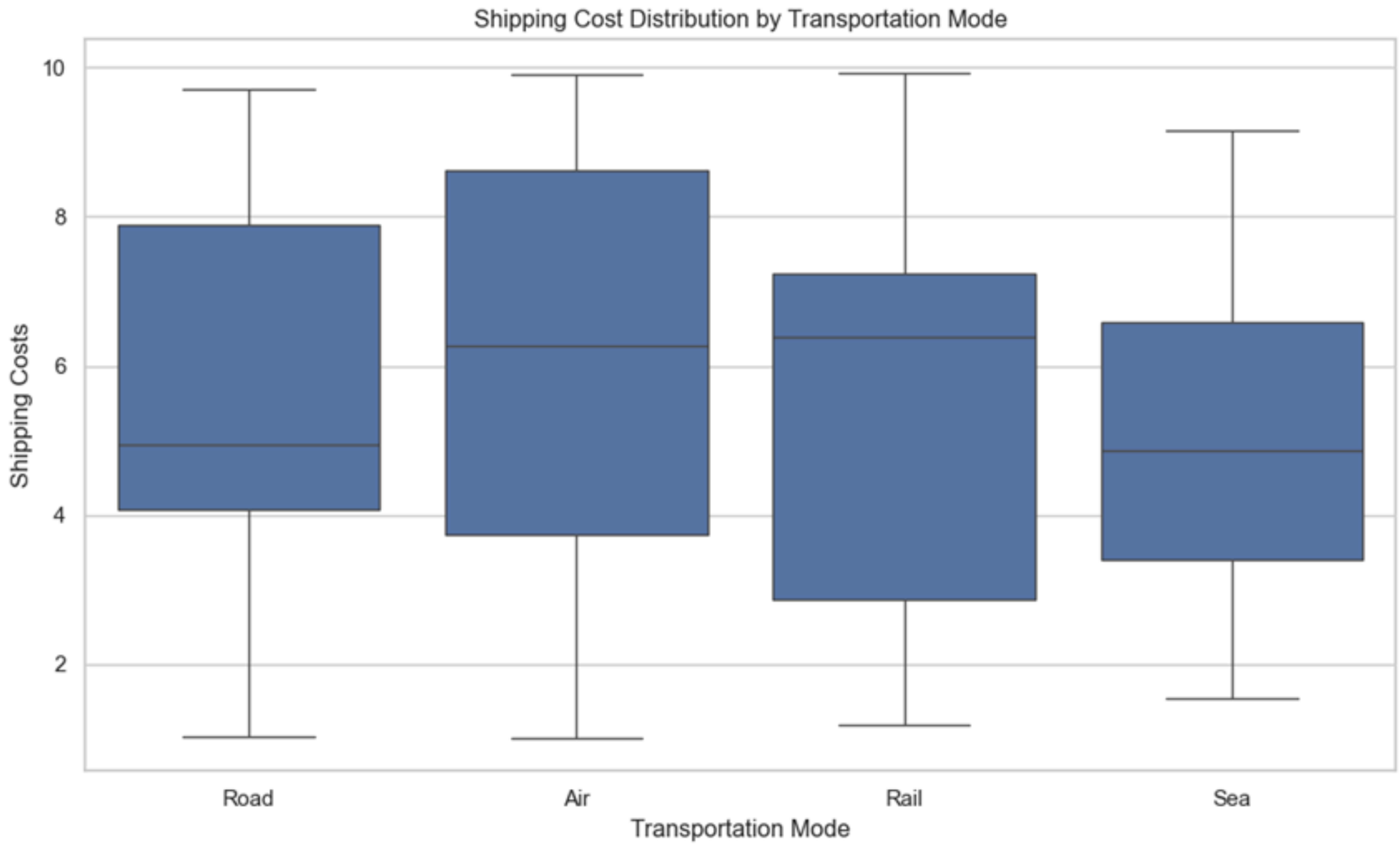
Insight:

- Sales show high volatility with frequent spikes and drops
- No clear upward or downward trend observed
- Demand influenced by external factors like seasonality

Business Value:

- Helps in understanding unstable demand patterns
- Supports better inventory and supply planning
- Enables proactive decision-making during demand fluctuations

SHIPPING COST BY TRANSPORTATION MODE



SHIPPING COST BY TRANSPORTATION MODE

Insight:

- Air transport shows higher median shipping cost
- Rail and Sea have moderate cost ranges
- Road transport has relatively lower and more stable costs

Business Value:

- Helps in selecting cost-efficient transportation modes
- Supports optimization of logistics and shipping strategies
- Enables balancing between cost and delivery speed

MACHINE LEARNING MODELS

Models used:

- Ridge Regression (baseline model)
- Random Forest (ensemble model)
- Extra Trees (advanced ensemble model)

Objective:

Predict product demand accurately

MACHINE LEARNING MODELS

Model Training Strategy

- Dataset split into training (70%) and testing (30%)
- Models trained on historical demand data
- Compared performance across models

Approach:

Select best model based on accuracy metrics

MODEL EVALUATION METRICS

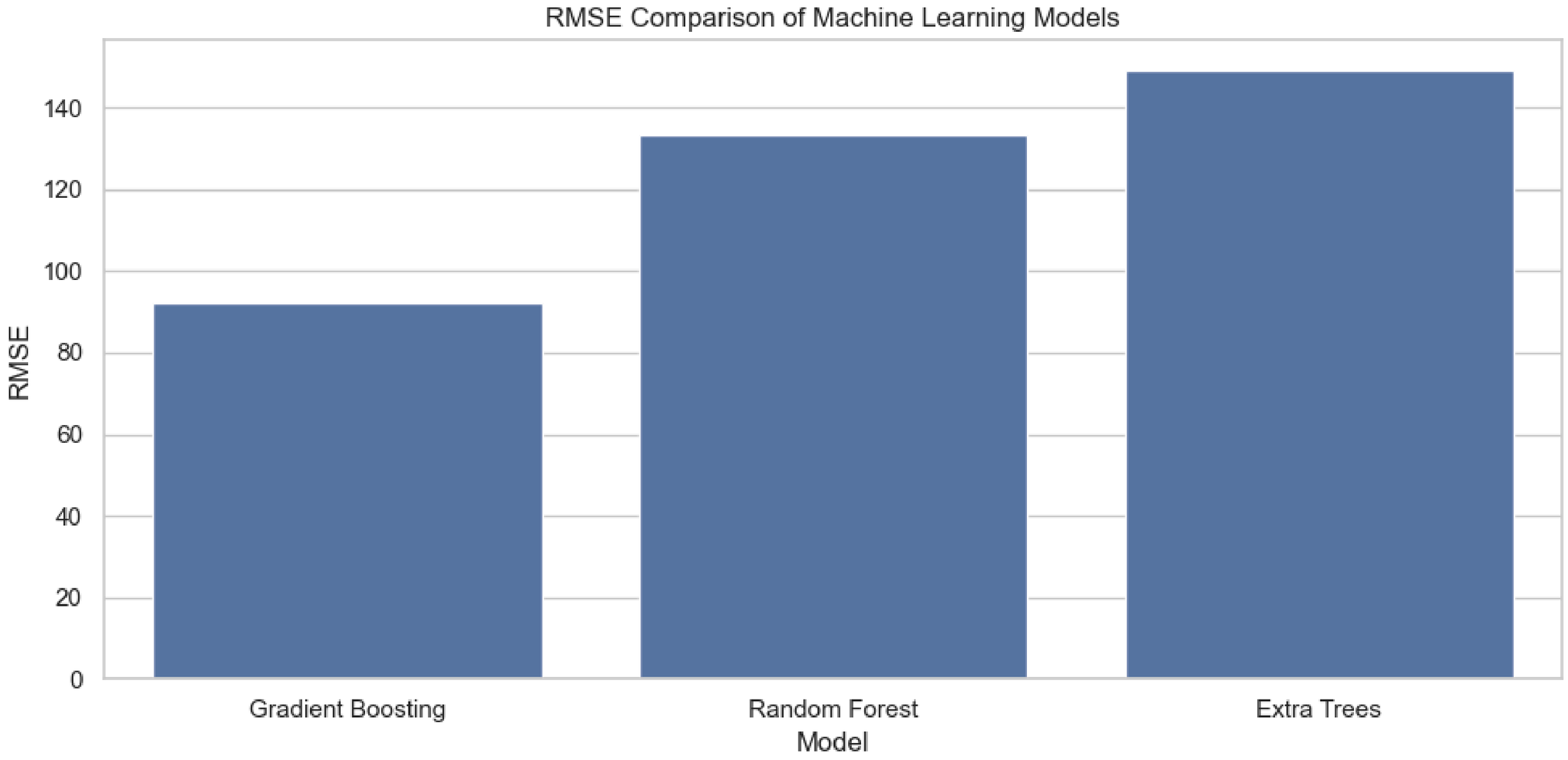
Used metrics:

- MAE → Average prediction error
- RMSE → Error sensitivity measure
- MAPE → Percentage error
- R^2 Score → Model accuracy

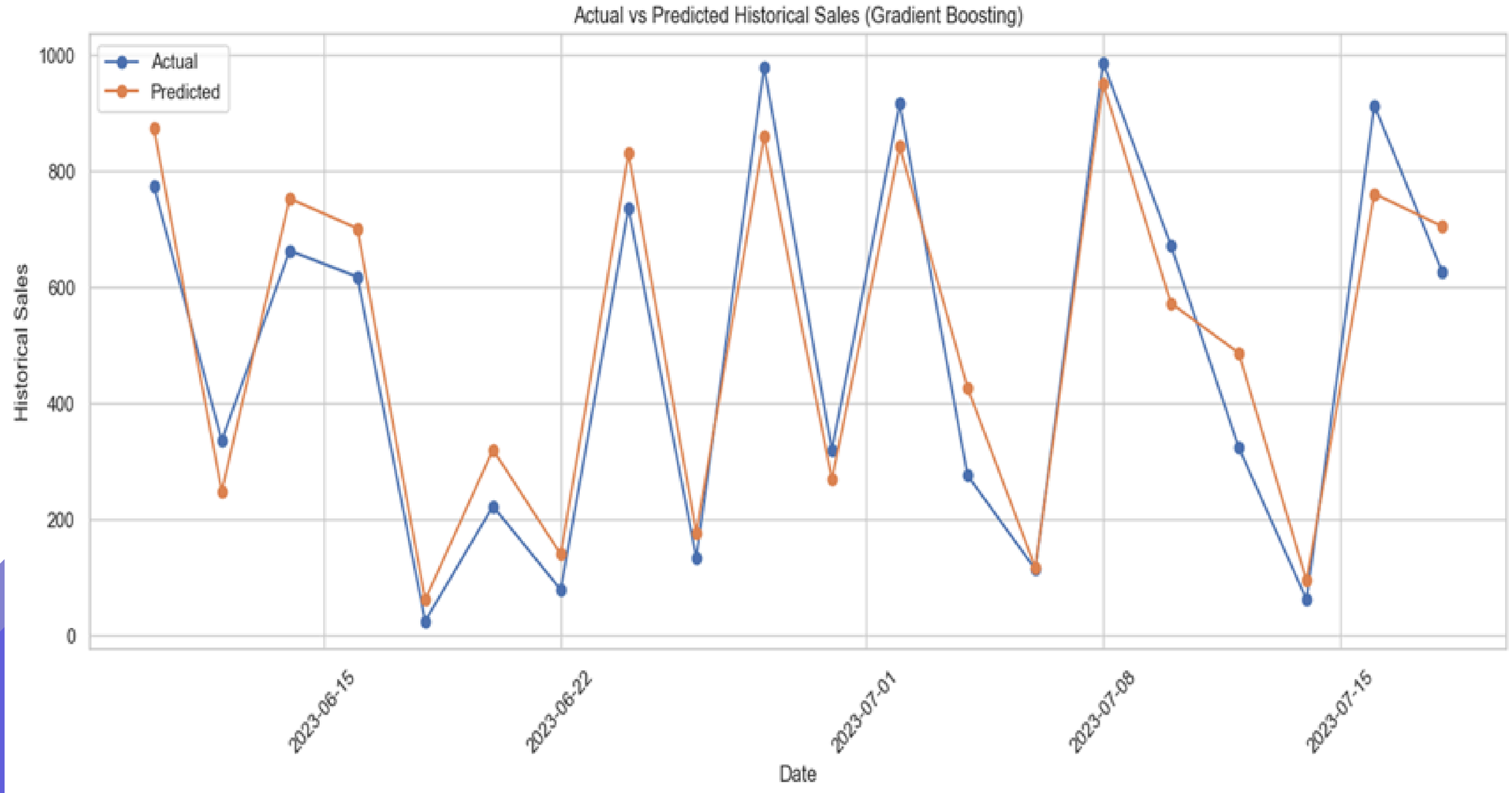
Insight:

- Ensemble models perform better than linear models
- Random Forest / Extra Trees give higher accuracy

MODEL PERFORMANCE



ACTUAL VS PREDICTED SALES



ACTUAL VS PREDICTED SALES

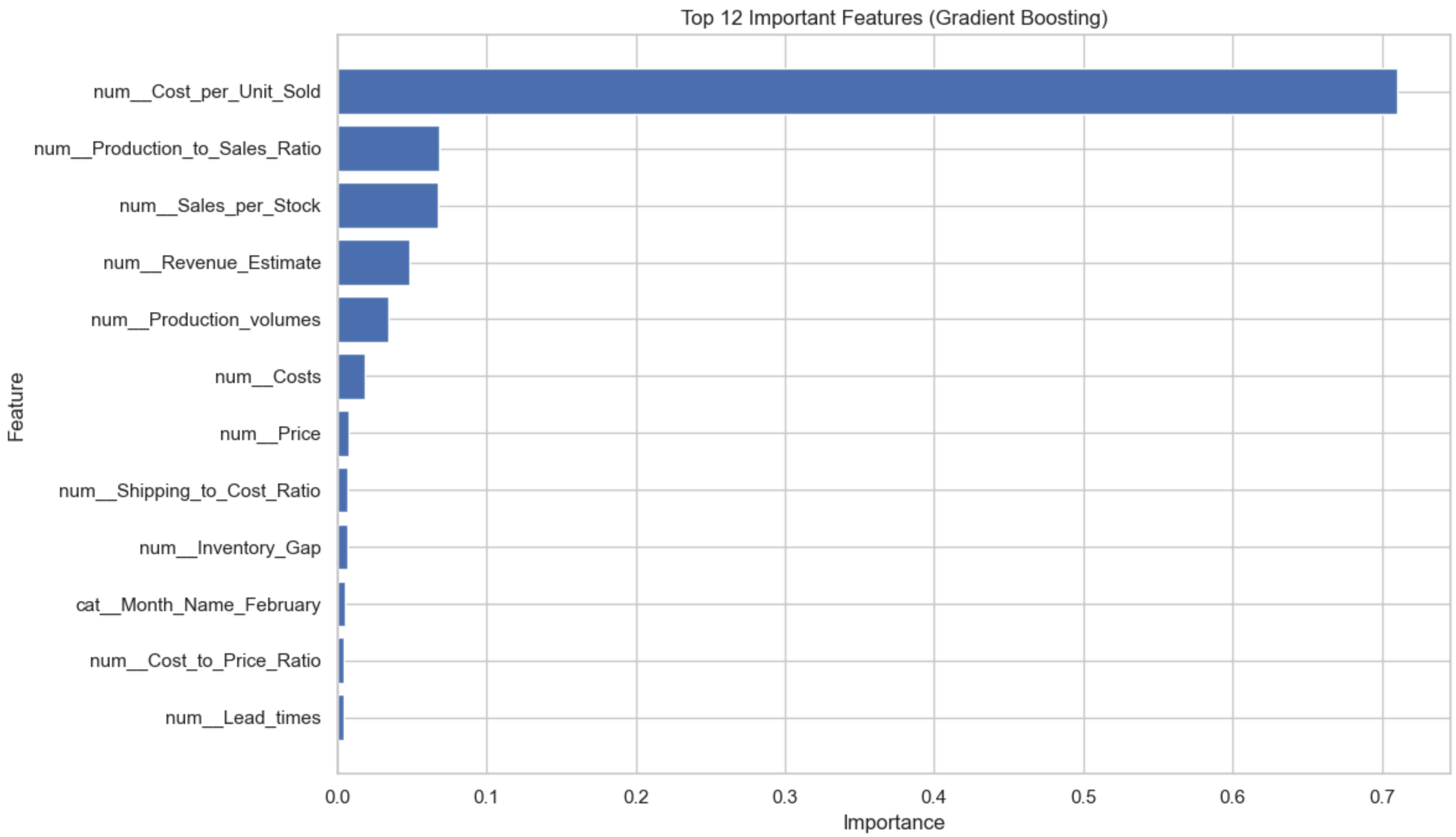
Insight:

- Predicted values closely follow actual demand
- Model captures demand trends effectively
- Minor deviations during extreme values

Business Value:

- Reliable demand forecasting
- Supports decision-making

FEATURE IMPORTANCE



PROJECT OUTCOME & FUTURE SCOPE

- Successfully analyzed product demand patterns
- Identified key high-demand products
- Built predictive ML model
- Generated actionable business insights
- Real-time demand forecasting system
- Integration with business dashboards
- Use of Deep Learning models
- Automated inventory management system

CONCLUSION

- Supply chain efficiency improves with ML
- Demand prediction helps reduce cost & risk
- Data-driven strategies enhance decision-making
- Project provides strong foundation for real-world applications

Thank You